Wave-Based EP Overlap

Pipeline MoE experts in waves inside one kernel

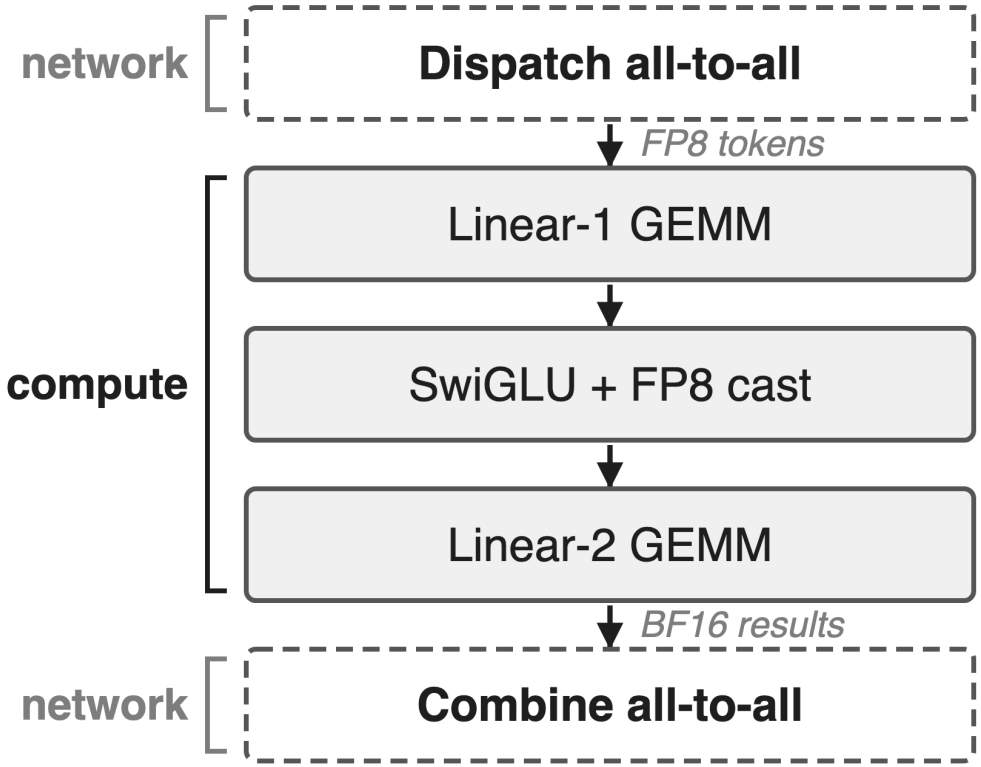
DeepSeek-AI · DeepSeek-V4

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Comm/Compute Overlap · arxiv.org/abs/2606.19348

Each MoE layer wraps GEMMs in two all-to-alls

- V4-Pro: **1.6T** params, **49B** active, **384** experts
- V4-Flash: **284B** / **13B**, **256** experts
- Each token picks **6** routed experts, mostly on other GPUs



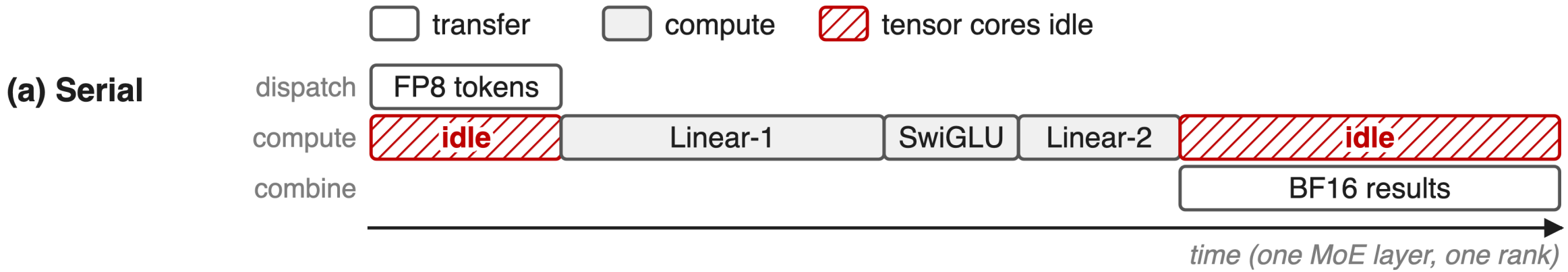
Communication costs as much as compute

≈ 1 : 1

computation-to-communication ratio of cross-node expert parallelism in DeepSeek-V3: run serially, a MoE layer waits on the network as long as it computes

DeepSeek-V3 technical report, arXiv 2412.19437, §3.2.1

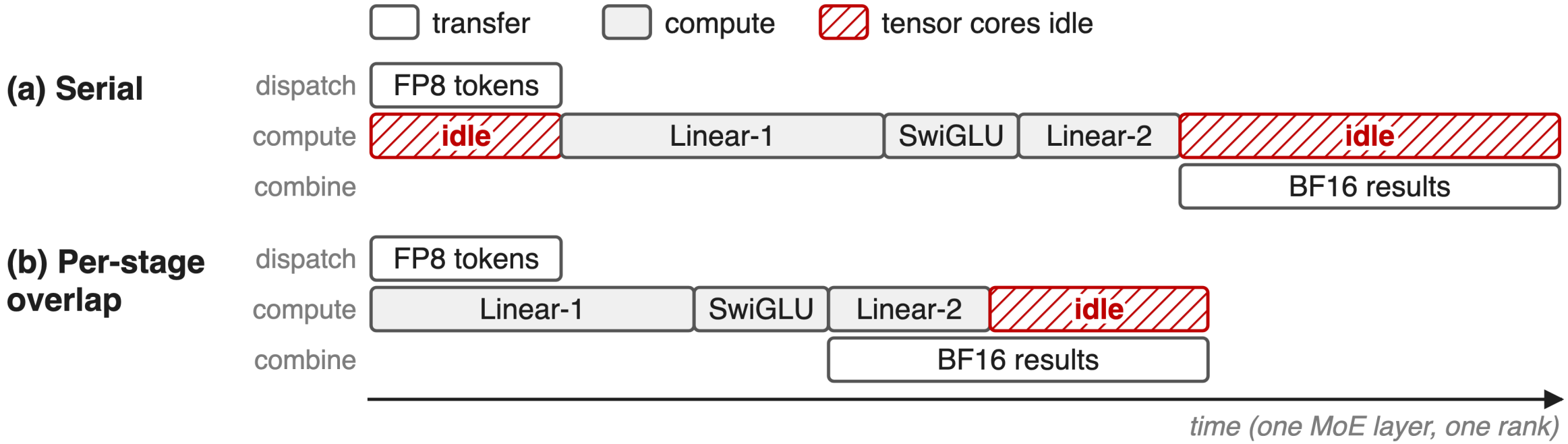
Serial stages leave tensor cores idle



Schematic timeline of one MoE layer on one rank; durations illustrative, not measured

Every GEMM waits for the whole dispatch; every result waits for the last GEMM

Per-stage overlap pairs mismatched stages



Batch-level overlap (V3 DualPipe, two micro-batches) needs a second, equal batch: see deck 11

Small FP8 dispatch hides under big Linear-1; big BF16 combine outlasts small Linear-2

Insight: communication is shorter than compute

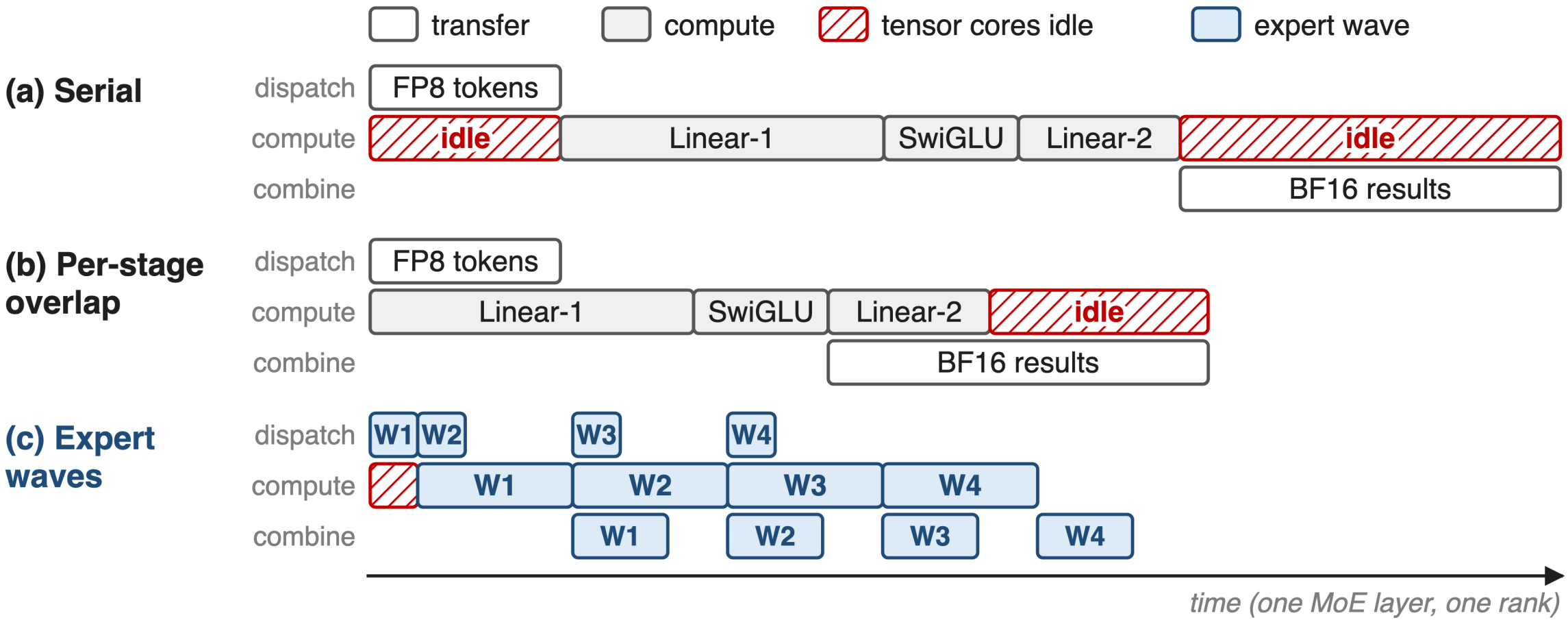
- Profiled V4 MoE layer: total transfer time **<** total compute time
- So the network is not the bound; the stage barriers are

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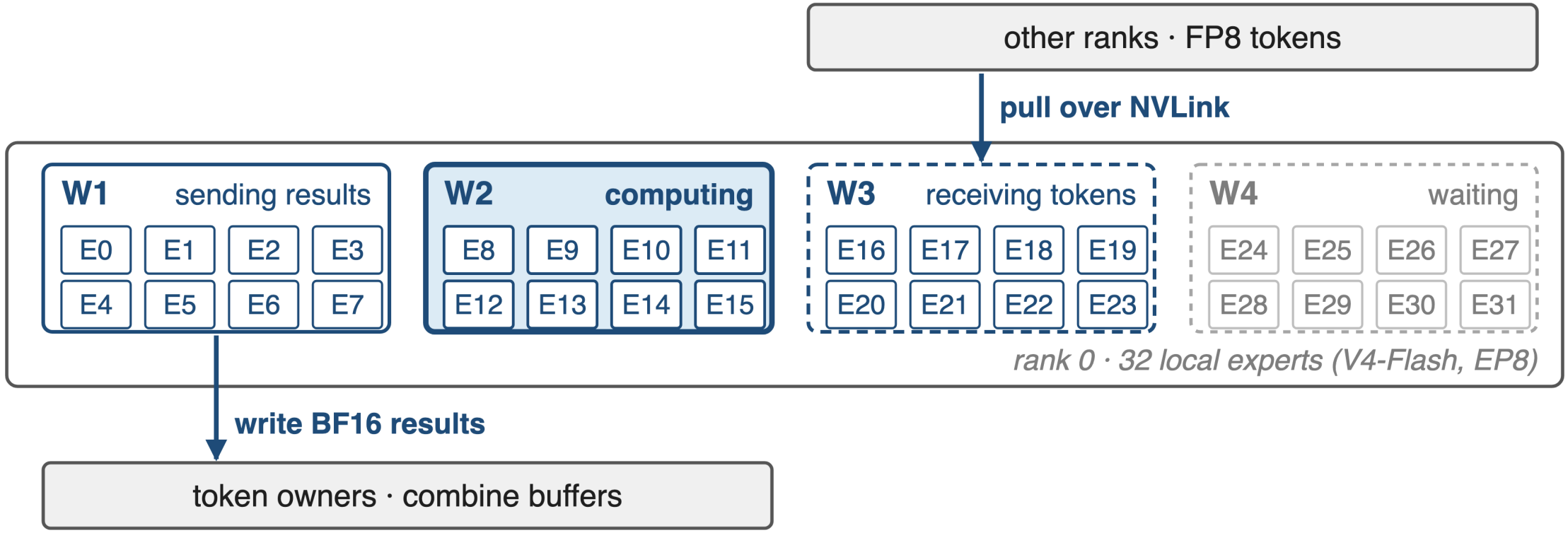
Pipeline at expert granularity and compute becomes the only bottleneck

Pipeline experts in waves inside one kernel



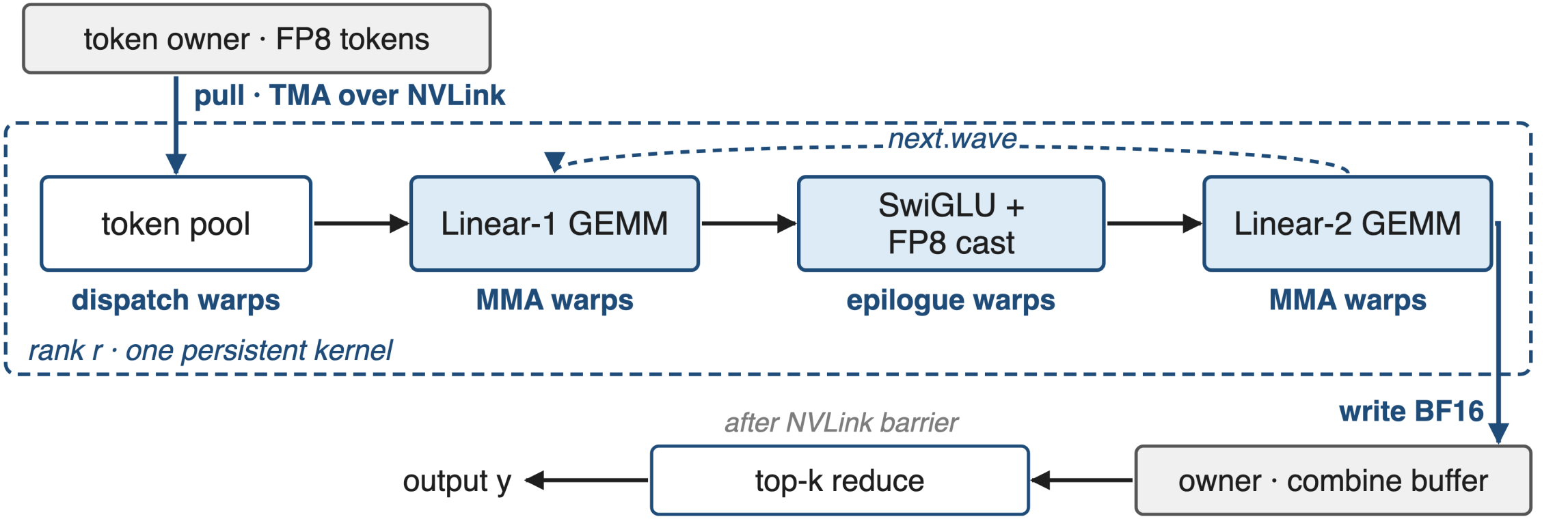
Schematic, 4 waves. Parts: ❶ expert waves, ❷ one fused pull-based kernel, ❸ wave sizing

❶ A wave computes as soon as its tokens land



Steady state: wave k computes while k+1 arrives and k−1 returns

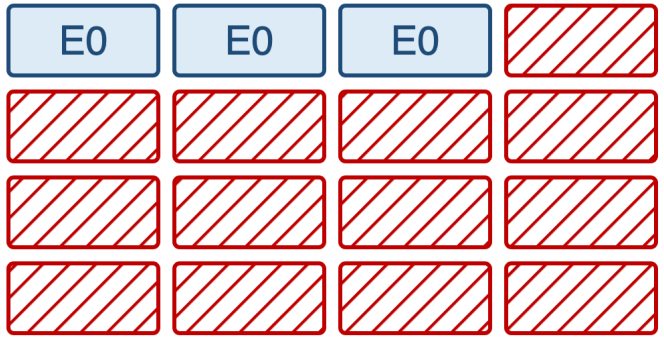
❷ One kernel with three warp roles



Dispatch pulls: no per-chunk notification latency as with fine-grained push

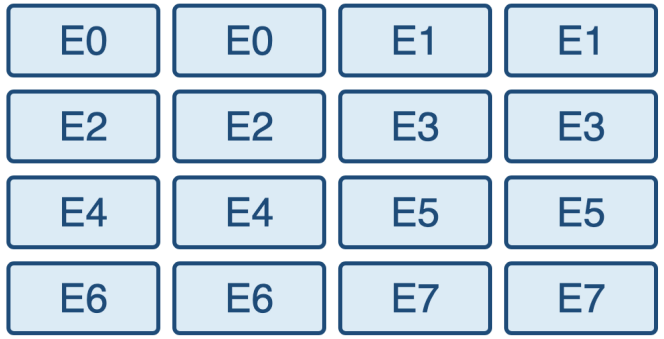
❸ Waves just big enough to fill every SM

1 expert per wave



too few tiles
SMs idle

8 experts per wave



smallest wave that
fills every SM

all 32 experts in one wave



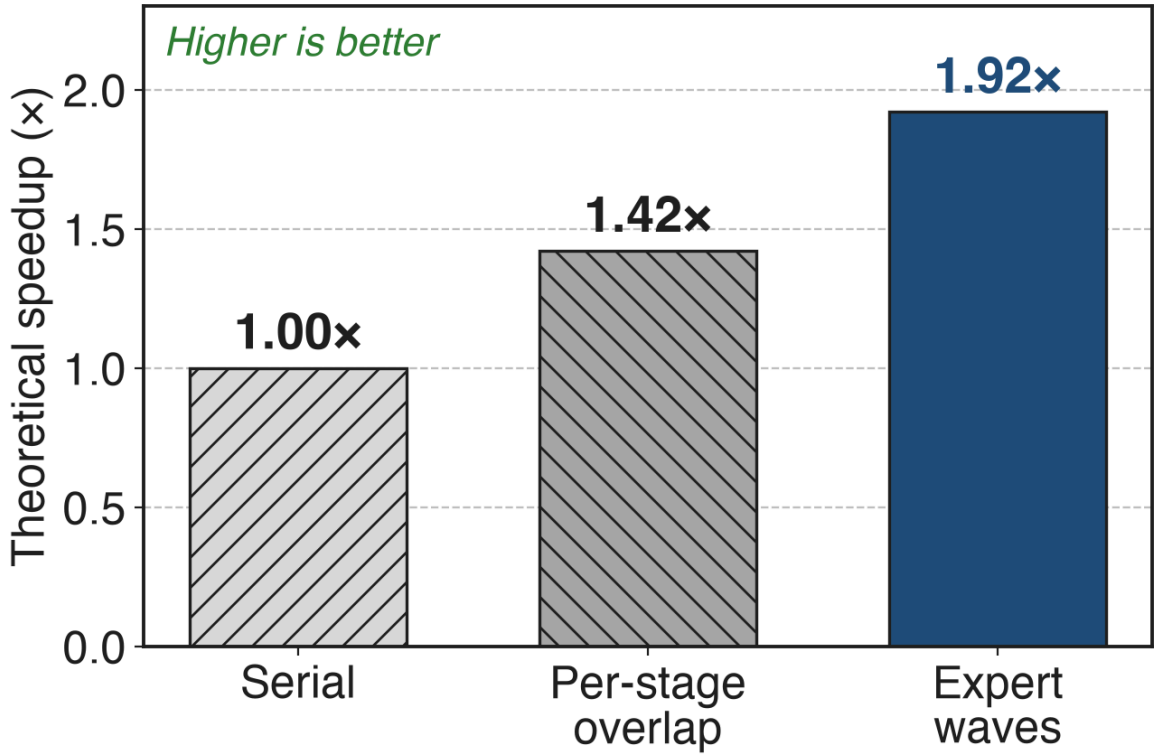
first GEMM waits for
every expert's tokens

Schematic: 16 SMs, one Linear-1 tile per cell. Smaller batches give fewer tiles per expert, so waves grow

Pick the fewest experts whose tiles cover 2× the SMs (at most 32)

Waves raise the overlap ceiling to 1.92×

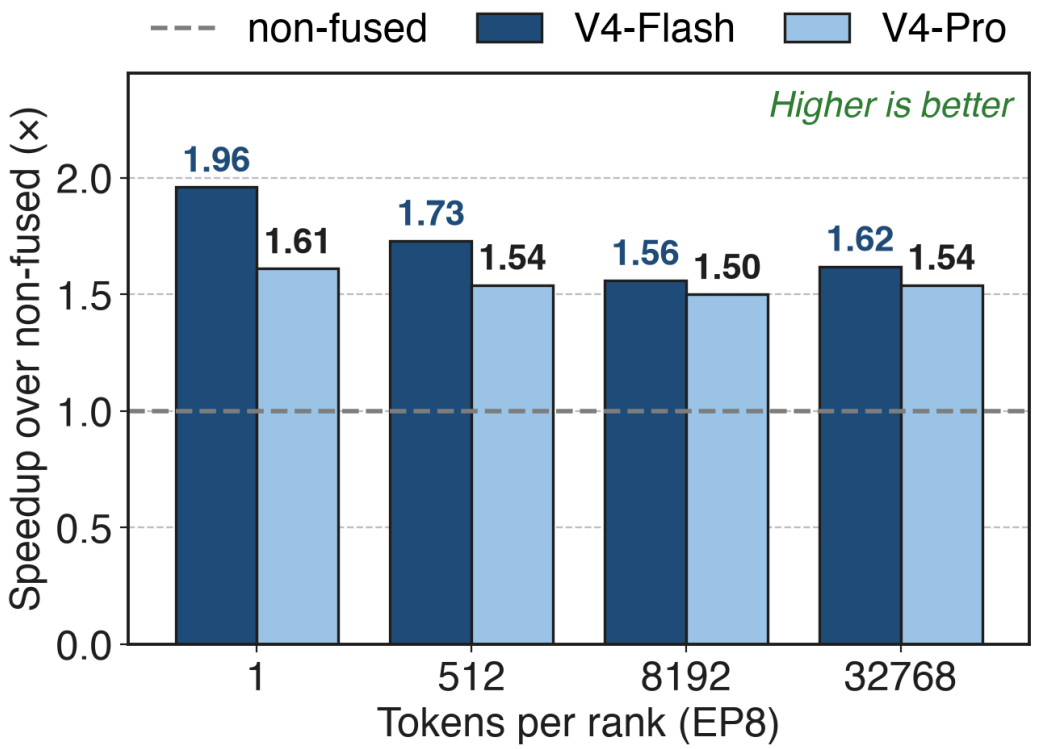
- V4-Flash configuration, the paper's model
- Per-stage overlap (Comet): **1.42×**
- Expert waves: **1.92×**



Theory assumes perfect overlap; fill and drain cost extra

Mega MoE runs 1.50–1.96× faster than non-fused

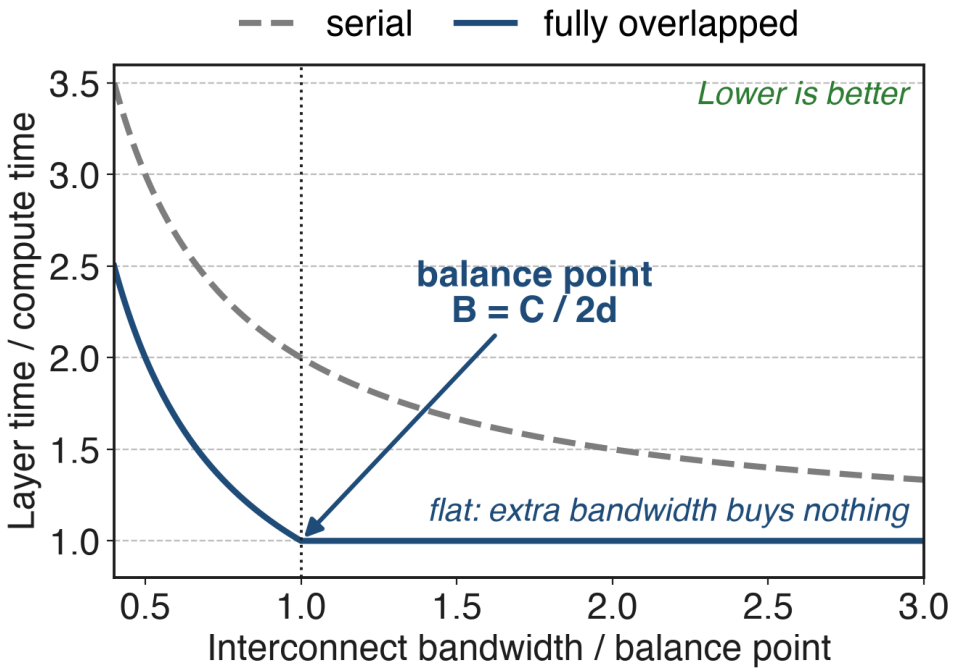
- One MoE layer, EP8, vs DeepEP + separate GEMMs
- Largest at **1** token per rank: rollout tail
- Also validated on Ascend NPUs



Tiny batches gain most: no second batch needed to hide transfers

Bandwidth beyond C/B = 2d buys nothing

- Per token-expert: **6hd** FLOPs, **3h** bytes
- V4-Pro: C/B ≤ 2d = **6144** FLOPs/Byte
- **1 GB/s** hides **6.1 TFLOP/s** of compute



Analytic, from DeepSeek-V4 §3.1; h = hidden size, d = expert width

Size interconnects to the balance point, not to the maximum

The cost: exposed edges, power, prerequisites

- **Fill and drain**: first dispatch, last combine stay exposed
- **Power**: compute, memory and network all run hot
- SwiGLU's exp and division stall the GEMM pipeline
- Needs symmetric memory, multi-process launch
- Release kernel: SM100, FP8×FP4 only
- "Still under development"

Summary

- Serial EP: transfers take as long as compute (≈ **1:1** in V3)
- **Insight**: a V4 layer communicates less than it computes
- Expert waves in one pull-based kernel
- **1.50–1.96×** over non-fused; balance point **2d** FLOPs/Byte

https://arxiv.org/abs/2606.19348